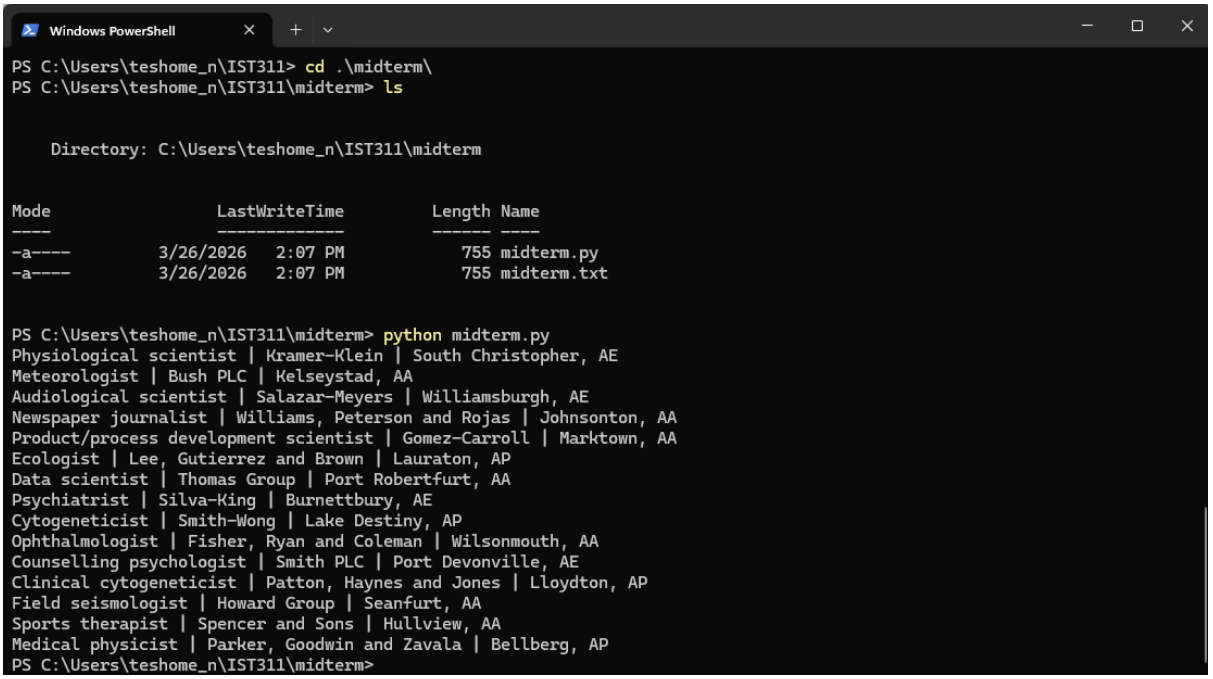


midterm.py midterm.txt

File Edit View

```
import requests
from bs4 import BeautifulSoup

url = "https://realpython.github.io/fake-jobs/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

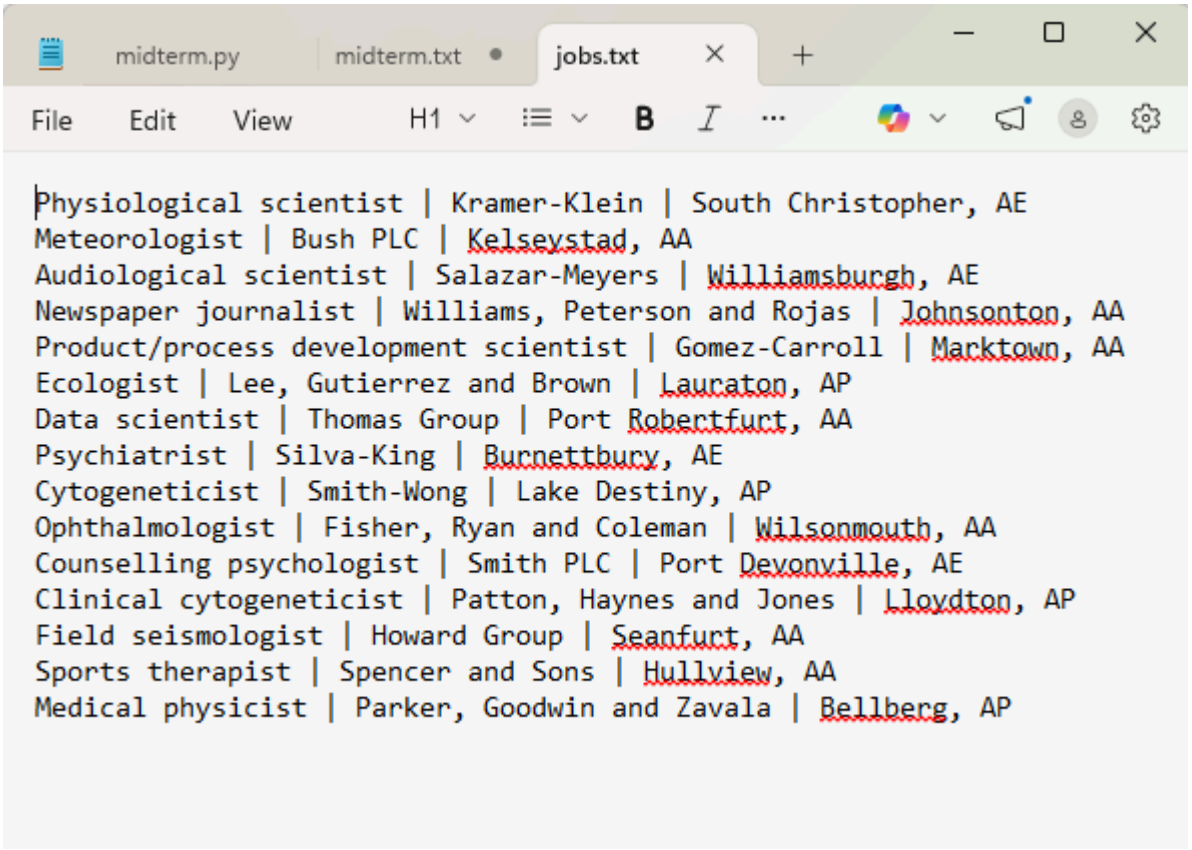
jobs = soup.find_all("div", class_="card-content")

results = []

for job in jobs:
    title = job.find("h2", class_="title").text.strip()
    company = job.find("h3", class_="company").text.strip()
    location = job.find("p", class_="location").text.strip()

    if title.lower().endswith("ist"):
        results.append((title, company, location))

with open("jobs.txt", "w") as f:
    for r in results:
        line = f"{r[0]} | {r[1]} | {r[2]}"
        print(line)
        f.write(line + "\n")
```


midterm.py
midterm.txt
jobs.txt
File
Edit
View
H1
Physiological scientist | Kramer-Klein | South Christopher, AE
Meteorologist | Bush PLC | Kelseystad, AA
Audiological scientist | Salazar-Meyers | Williamsburgh, AE
Newspaper journalist | Williams, Peterson and Rojas | Johnsonton, AA
Product/process development scientist | Gomez-Carroll | Marktown, AA
Ecologist | Lee, Gutierrez and Brown | Lauraton, AP
Data scientist | Thomas Group | Port Robertfurt, AA
Psychiatrist | Silva-King | Burnettbury, AE
Cytogeneticist | Smith-Wong | Lake Destiny, AP
Ophthalmologist | Fisher, Ryan and Coleman | Wilsonmouth, AA
Counselling psychologist | Smith PLC | Port Devonville, AE
Clinical cytogeneticist | Patton, Haynes and Jones | Lloydton, AP
Field seismologist | Howard Group | Seanfurt, AA
Sports therapist | Spencer and Sons | Hullview, AA
Medical physicist | Parker, Goodwin and Zavala | Bellberg, AP

SQLQuery1.sql - csd...1Projs (IST311 (71))*

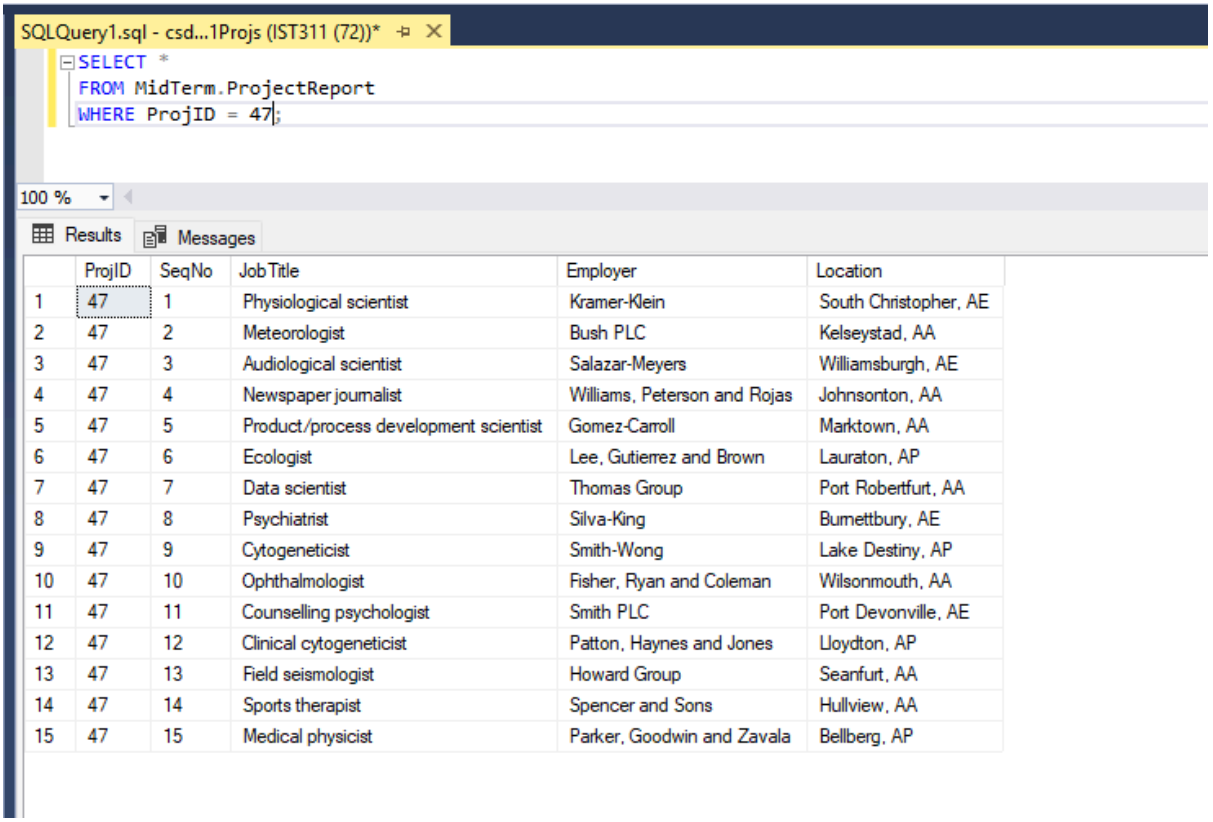
```
SELECT ProjID, MUID, FName, LName
From Midterm.ProjSubmission
```

100 %

Results | Messages

	ProjID	MUID	FName	LName
1	1	8	Nadia	Gadson
2	2	8	Conner	Kahn
3	3	98	Conner	Kahn
4	4	5	Jaylene	Bayona
5	5	53	Matthew	Padua
6	6	30	Jace	Higdon
7	7	1	Djouvens	Narcisse
8	8	78	Samantha	Butler
9	9	25	Katrina	Hauser
10	10	25	Katrina	Hauser
11	11	77	Avery	Dunn
12	12	66	Brett	Cannon
13	13	6	Ryan	Pitts
14	14	11050078	Kalyn	LeBlanc
15	15	11050078	Kalyn	LeBlanc
16	16	11050078	Kalyn	LeBlanc
17	17	11050078	Kalyn	LeBlanc
18	18	81	Amen	Gando...
19	19	11069794	Camauri	Simmons
20	20	11069794	Camauri	Simmons
21	21	5	Zaccary	Priest
22	22	18	Parker	Hodges
23	23	15	Kabir	Singh
24	24	15	Kabir	Singh
25	31	82	Cristian	Martin...
26	44	53	Mike	Kwan
27	45	53	Mike	Kwan
28	47	1	Nathan	Tesho...

Query executed successfully.



	ProjID	SeqNo	JobTitle	Employer	Location
1	47	1	Physiological scientist	Kramer-Klein	South Christopher, AE
2	47	2	Meteorologist	Bush PLC	Kelseystad, AA
3	47	3	Audiological scientist	Salazar-Meyers	Williamsburgh, AE
4	47	4	Newspaper journalist	Williams, Peterson and Rojas	Johnsonton, AA
5	47	5	Product/process development scientist	Gomez-Carroll	Marktown, AA
6	47	6	Ecologist	Lee, Gutierrez and Brown	Lauraton, AP
7	47	7	Data scientist	Thomas Group	Port Robertfurt, AA
8	47	8	Psychiatrist	Silva-King	Burnettbury, AE
9	47	9	Cytogeneticist	Smith-Wong	Lake Destiny, AP
10	47	10	Ophthalmologist	Fisher, Ryan and Coleman	Wilsonmouth, AA
11	47	11	Counselling psychologist	Smith PLC	Port Devonville, AE
12	47	12	Clinical cytogeneticist	Patton, Haynes and Jones	Lloydton, AP
13	47	13	Field seismologist	Howard Group	Seanfurt, AA
14	47	14	Sports therapist	Spencer and Sons	Hullview, AA
15	47	15	Medical physicist	Parker, Goodwin and Zavala	Bellberg, AP

Report:

This project was completed using Python with the requests, BeautifulSoup, and pyodbc libraries. PowerShell was used to run scripts, and SQL Server Management Studio was used to verify database records. A Python script was used to scrape job listings from a provided website. The HTML was parsed using BeautifulSoup, and job data (title, employer, and location) was extracted. Only jobs matching the required condition (titles ending in “ist”) were kept. The results were saved to a text file. The filtered job data was inserted into a SQL Server database using pyodbc. A ProjID was created first, then each job record was inserted with a sequence number. The data was verified in SSMS to confirm successful insertion. The script successfully scraped, filtered, and stored job data in both a text file and SQL database. Output was consistent across both formats.This project demonstrated basic web scraping and database integration using Python, including data extraction, filtering, file handling, and SQL insertion.

Python Script:

```
import requests
from bs4 import BeautifulSoup

url = "https://realpython.github.io/fake-jobs/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

jobs = soup.find_all("div", class_="card-content")

results = []

for job in jobs:
    title = job.find("h2", class_="title").text.strip()
    company = job.find("h3", class_="company").text.strip()
    location = job.find("p", class_="location").text.strip()

    if title.lower().endswith("ist"):
        results.append((title, company, location))

# Save to file
with open("jobs.txt", "w") as f:
    for r in results:
        line = f"{r[0]} | {r[1]} | {r[2]}"
        print(line)
        f.write(line + "\n")

import pyodbc

conn = pyodbc.connect(
    "DRIVER={SQL Server};"
    "SERVER=csdata.cd4sevot432y.us-east-1.rds.amazonaws.com;"
    "DATABASE=IST311Projs;"
    "UID=IST311;"
    "PWD=i$t311spR26"
)

cursor = conn.cursor()

proj_id = 47

seq = 1

for r in results:
    cursor.execute("""
        INSERT INTO MidTerm.ProjectReport
        VALUES (?, ?, ?, ?, ?)
    """, proj_id, seq, r[0], r[1], r[2])
```

```
    seq += 1

conn.commit()
conn.close()
```

Output:
Physiological scientist | Kramer-Klein | South Christopher, AE
Meteorologist | Bush PLC | Kelseystad, AA
Audiological scientist | Salazar-Meyers | Williamsburgh, AE
Newspaper journalist | Williams, Peterson and Rojas | Johnsonton, AA
Product/process development scientist | Gomez-Carroll | Marktown, AA
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